

# Concurrent Programming Keywords in Java and Rust

## Java Concurrent Programming

Threading and Execution:

- Thread: Class to create and manage threads
- Runnable: Functional interface for a thread task
- Callable: Like Runnable, but returns a value
- Executor, ExecutorService: Manage thread pools
- Future, CompletableFuture: Async task results and chaining

Synchronization:

- synchronized: Locking methods or blocks
- volatile: Ensures visibility across threads
- wait()/notify(): Coordination between threads
- Lock, ReentrantLock, Condition: Explicit locks and conditions

Concurrency Utilities:

- ConcurrentHashMap, CopyOnWriteArrayList
- AtomicInteger, etc.: Lock-free atomics
- CountdownLatch, CyclicBarrier, Semaphore
- BlockingQueue: Safe producer/consumer communication
- ForkJoinPool: Work-stealing parallelism

## Rust Concurrent Programming

Rust Threading:

- thread::spawn: Spawns a new thread
- JoinHandle: Returned by spawn, used to join thread
- move: Closure takes ownership of captured variables

Channels & Message Passing:

- std::sync::mpsc: Multi-producer, single-consumer channels
- mpsc::channel(): Create Sender/Receiver
- send()/recv(): Communication methods

Synchronization:

- Mutex, RwLock: Thread-safe shared state
- Arc<T>: Thread-safe reference counting
- Condvar: Conditional variable
- Atomic types: AtomicBool, AtomicUsize, etc.

Async/Await:

- async, await: Declare and wait for async code
- Future: Trait for async values
- tokio, async-std: Async runtimes

## Summary

Summary Table:

| Feature         | Java                        | Rust                         |  |
|-----------------|-----------------------------|------------------------------|--|
| Threads         | Thread, Runnable            | thread::spawn, JoinHandle    |  |
| Async Tasks     | CompletableFuture           | async, await, Future, tokio  |  |
| Locks           | synchronized, ReentrantLock | Mutex, RwLock                |  |
| Atomics         | AtomicInteger, volatile     | AtomicUsize, AtomicBool      |  |
| Message Passing | BlockingQueue, Executor     | mpsc::channel, send()/recv() |  |
| Sync Mechanism  | CountDownLatch, Semaphore   | Condvar, channels            |  |
| Smart Pointers  | N/A                         | Arc<T>, Rc<T>                |  |